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D

Each emergent light beam has the amplitude $A' = A \cos 45^\circ$

As there are two incident beams incident on the polarizer in phase, due to superposition the total emergent light beam has amplitude $= 2A' = 2A \cos 45^\circ = \sqrt{2}A$

Since $I \propto A^2$

$$\frac{I_E}{I} = \left(\frac{\sqrt{2}A}{A} \right)^2 = 2$$

$$I_E = 2I$$

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C



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